

Bill of Materials - ACT-PKR-4500 'Patriot' Retrievable Packer, 4-1/2 in.

Engineering Bill of Materials | Document BOM-PKR-4500 Rev C

Product / Assembly	ACT-PKR-4500
Document No. / Rev	BOM-PKR-4500 Rev C
Effective Date	2024-10-15
Prepared By	A. Gutierrez
Checked By	T. Okafor
Approved By	J. Martinez, P.E.
ERP Reference	ITEM-PKR-4500

Revision History

Rev	Date	Description of Change	By
A	2023-06-12	Initial release	R. Callahan
B	2024-02-20	Slip retainer screw upgraded to Gr 8; torque note added	A. Gutierrez
C	2024-10-15	Added optional Aflas element kit -03-AF per ECO-2024-067	R. Callahan

Bill of Materials

Hierarchy: Level 0 = top assembly; Level 1 = component or sub-assembly; Level 2 = sub-assembly component. Indented descriptions belong to the sub-assembly immediately above. Quantities are per one (1) next-higher assembly.

Item	Lvl	Part Number	Description	Qty	UOM	Material / Spec	Notes
1	0	ACT-PKR-4500	PACKER ASSY, 4-1/2 IN. HYDRAULIC SET RETRIEVABLE, 10K	1	EA	-	Top-level assembly
2	1	ACT-PKR-4500-01	■ SUB, TOP, 2-7/8 EUE BOX UP	1	EA	AISI 4140 HT 28-32 HRC	Dwg D-PKR-4500-01 Rev C
3	1	ACT-PKR-4500-02	■ MANDREL, PACKER, 4-1/2	1	EA	AISI 4140 HT 28-32 HRC	Heat-traceable; Dwg D-PKR-4500-02 Rev C
4	1	ACT-PKR-4500-03	■ ELEMENT ASSY, 3-PIECE	1	EA	-	See sub-items; opt. Aflas kit -03-AF per ECO-2024-067
5	2	ACT-PKR-4500-03A	■ RING, END, ELEMENT (ANTI-EXTRUSION)	2	EA	AISI 1018, BRONZE BACKUP	
6	2	ACT-PKR-4500-03B	■ ELEMENT, CENTER, 90 DURO	1	EA	HNBR 90A (CPD ACT-H90)	Std; Aflas 90A for H2S >10% (kit -03-AF)
7	2	ACT-PKR-4500-03C	■ ELEMENT, END, 80 DURO	2	EA	HNBR 80A (CPD ACT-H80)	
8	2	ACT-PKR-4500-03D	■ SPACER RING, ELEMENT	2	EA	AISI 1026	

Item	Lvl	Part Number	Description	Qty	UOM	Material / Spec	Notes
9	1	ACT-PKR-4500-04	■ SLIP ASSY, UPPER	1	EA	-	
10	2	ACT-PKR-4500-04A	■ CONE, UPPER SLIP	1	EA	AISI 4140 HT 28-32 HRC	
11	2	ACT-PKR-4500-04B	■ SLIP, SEGMENT, WICKER TYPE	6	EA	AISI 8620 CARBURIZED 58-62 HRC	Case depth 0.030-0.045 in.
12	2	ACT-PKR-4500-04C	■ RETAINER, SLIP, BAND TYPE	1	EA	AISI 1018	
13	2	ACT-PKR-4500-04D	■ SCREW, SLIP RETAINER, 1/4-20 UNC	6	EA	ALLOY STEEL GR 8	Torque per TS-100 Table 6
14	1	ACT-PKR-4500-05	■ SLIP ASSY, LOWER	1	EA	-	Mirror of item -04
15	2	ACT-PKR-4500-05A	■ CONE, LOWER SLIP	1	EA	AISI 4140 HT 28-32 HRC	
16	2	ACT-PKR-4500-05B	■ SLIP, SEGMENT, WICKER TYPE	6	EA	AISI 8620 CARBURIZED 58-62 HRC	
17	2	ACT-PKR-4500-05C	■ RETAINER, SLIP, BAND TYPE	1	EA	AISI 1018	
18	1	ACT-PKR-4500-06	■ PISTON, HYDRAULIC SETTING	1	EA	AISI 4140 HT 28-32 HRC	
19	1	ACT-PKR-4500-07	■ RING, SHEAR, SETTING (2,950 +/- 150 PSI)	1	EA	AISI 1144	Do NOT substitute; sets shear-out pressure
20	1	ACT-PKR-4500-08	■ LOCK RING, BODY, RATCHET	1	EA	AISI 4140 HT 32-36 HRC	
21	1	ACT-PKR-4500-09	■ SUB, BOTTOM, 2-7/8 EUE PIN DOWN	1	EA	AISI 4140 HT 28-32 HRC	
22	1	ACT-PKR-4500-10	■ SCREW, SET, DRAG BLOCK, 3/8-16 UNC	8	EA	ALLOY STEEL GR 8	Torque 25 ft-lb per TS-100
23	1	ACT-PKR-4500-RK	■ REDRESS KIT, PKR-4500 (SEAL & SHEAR)	1	KT	-	See sub-items; order separately
24	2	ACT-ORG-228-V75	■ O-RING, -228, 75 DURO	4	EA	VITON 75A	
25	2	ACT-ORG-330-V75	■ O-RING, -330, 75 DURO	2	EA	VITON 75A	
26	2	ACT-BU-330-P	■ BACK-UP RING, -330, PEEK	4	EA	PEEK	
27	2	ACT-PKR-4500-07	■ RING, SHEAR, SETTING	1	EA	AISI 1144	Same as item 20

General Notes

- All elastomer components are single-use: replace at every redress from the applicable redress kit.
- Load-path metallic components require heat traceability to a mill certificate (EN 10204 Type 3.1) retained in the device history record.
- Shear devices (rings, screws, pins) are calibrated items - substitute parts are prohibited.
- Torque all threaded connections per ACT specification TS-100 (packers/plugs) or TS-200 (accessories).
- Sour service (NACE MR0175/ISO 15156) configurations use the materials noted; verify trim level on the sales order before assembly.
- Referenced documents: datasheet DS-PKR-4500, applicable manual/procedure, and quality plan ACT-QP-08.

Approvals

Role	Name	Signature	Date
Design Engineering	A. Gutierrez	/signed electronically/	2024-10-15
Manufacturing Engineering	T. Okafor	/signed electronically/	2024-10-15
Quality Assurance	S. Nguyen	/signed electronically/	2024-10-15
Engineering Manager	J. Martinez, P.E.	/signed electronically/	2024-10-15